

Candidate Name: _____

Class

Adm. No.

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2009 Promotional Examination 2

Pre-University 2

MATHEMATICS

Higher 1

8863/01

THURSDAY

17 SEPTEMBER

3 hours

Additional materials:

Cover Page

Writing papers

List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name, admission number and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

Section A: Pure Mathematics [40 marks]

- 1 Determine the range of values of m for which the curve $y = 2x^2 + m$ does not meet the line $y = (m - 3)x + 3$. [3]

- 2 Prove that the normal at $P(a, 2a)$ on the curve $y = 2\sqrt{ax}$ has the equation $y + x = 3a$. [3]

- 3 Solve the following simultaneous equations

$$2^{3x-6} = \frac{1}{8^y},$$

$$\lg y = 2\lg x + \lg 3. \quad [4]$$

- 4 (a) Without the use of graphic calculator, find the exact value of $\int_1^3 \frac{1}{3x-2} dx$. [2]

- (b) Find the area enclosed by the curve $y = \sqrt{5-x}$, its tangent at the point $(1, 2)$ with equation $4y + x = 9$ and the x -axis, giving your answer to 2 decimal places. [3]

- 5 Sketch, on the same diagram, stating clearly the equations of any asymptotes and the coordinates of any intersections with the axes, the graphs of

(i) $y = \sin 2x + 1$ for $0 \leq x \leq 2\pi$, [2]

(ii) $y = \ln(x-1)$ for $0 \leq x \leq 2\pi$. [2]

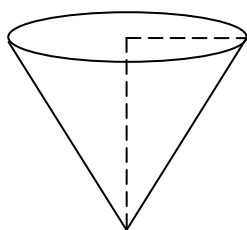
Hence, solve the inequality $\sin 2x + 1 \geq \ln(x-1)$ and state the number of roots of the equation $\sin 2x + 1 = |\ln(x-1)|$ for $0 \leq x \leq 2\pi$. [3]

- 6 (a) Differentiate each of the following expressions with respect to x , simplifying your answers:

(i) $y = (kx - 3)^5$, k is a constant, [2]

(ii) $y = e^{5x-x^2}$. [2]

(b)



A hollow right circular cone, initially empty, is being filled with water at the rate of $4\pi \text{ cm}^3 \text{ s}^{-1}$. At the same time, it is leaking at the rate of $2\pi \text{ cm}^3 \text{ s}^{-1}$. The height of the cone is twice its radius. Find, when the volume of the water in the cone is $\frac{128\pi}{3} \text{ cm}^3$,

(i) the radius of the water surface in the cone, [3]

(ii) the rate at which the radius of the water surface is changing at this instant. [2]

[** Volume of cone = $\frac{1}{3} \times \text{Base Area} \times \text{Height}$.]

7 The functions f and g are defined as

$$f : x \rightarrow (2x - 3)^2 - 1, \quad x \in \mathbb{R} \quad \text{and}$$

$$g : x \rightarrow e^{2x}, \quad x \in \mathbb{R}, \quad x \geq 0.$$

- (i) State with a reason which of the inverse functions, f^{-1} or g^{-1} does not exist. [2]
- (ii) Define the inverse function which exists. [3]
- (iii) Determine, with a reason, whether the composite function $f \circ g$ exists. If the function exists, give its rule and its domain. [4]

Section B: Statistics [60 marks]

8 During a recent economic crisis, the applicants for subsidised food coupons at a community centre come from 4 districts within the community. The number of applicants from each district is as follows:

District	Number of Applicants
A	320
B	210
C	220
D	250

If a survey on the food coupons is to be conducted, describe how a sample group of size 100 can be obtained by

- (i) simple random sampling,
 - (ii) stratified random sampling.
- Give an advantage for each sampling method. [7]

9 Athletes who are training for a particular race need to complete both the running and cycling segments. From past records, the time needed, in minutes, by athlete A and athlete B to complete each of the segments follows a normal distribution with mean and standard deviation as shown in the table below:

	Run Segment (in min)		Cycling Segment (in min)	
	Mean	Standard deviation	Mean	Standard deviation
Athlete A	50	12	28	4
Athlete B	58	10	25	3

Find the probability that

- (i) athlete A will complete the run segment of a particular race in less than 48 min, [2]
- (ii) athlete B will finish a particular race within 80 min, [3]
- (iii) athlete A will finish a particular race before athlete B by at least 10 min. [4]

10 A publishing company found errors in 3% of the total number of pages published by the company. Calculate the probability that

- (i) a 20-page magazine published by the company contains no error, [2]
- (ii) out of 12 issues of the magazine, there are more than 5 issues with no error. [3]

Using a suitable approximation, estimate the probability that out of 300 issues of the magazine published by the company, less than 155 issues have no error. [5]

- 11** The XYZ Airline has flights departing to several destinations from Singapore Changi airport. The table below shows the distance, d km, of some of these destinations from Singapore and their corresponding airfare, f .

Distance, d (km)	380	612	1017	1555	1358	630
Airfare, f (\$)	230	295	750	1300	760	500

- (i) Give a sketch of the scatter diagram for the data. [2]
 - (ii) Calculate the product moment correlation coefficient of the scatter data. Comment on the value obtained. [2]
 - (iii) Calculate the equation of the regression line of f on d . [1]
 - (iv) Calculate the equation of the regression line of d on f . [1]
 - (v) Use the appropriate regression line to estimate, to the nearest whole number,
 - (a) the airfare of a city which is 2283 km from Singapore, [1]
 - (b) the distance of a city from Singapore if the airfare is priced at \$600. [1]
 - (vi) Comment briefly on the reliability of the estimates in part (v). [2]
- 12** (a) In the year 2025, a population is hit by 2 new strains of flu virus. On average, out of 1000 people, 175 are infected by strain X, 160 are infected by strain Y and 35 are infected by both of the new strains. Find the probability that a randomly chosen person is
- (i) infected by strain X or strain Y (or both) of the flu virus, [2]
 - (ii) infected by both strains of flu, given that he is infected by strain X. [2]
- (b) In a certain sample space, the events A and B are such that $P(A) = 0.5$, $P(A \cup B)' = 0.2$ and $P(B|A) = 0.4$. Find
- (i) $P(A \cup B)$, [1]
 - (ii) $P(A \cap B)$, [2]
 - (iii) $P(B' \cap A)$. [2]
- State, giving a reason, whether events A and B are independent. [2]

- 13 A food company is preparing to launch the sales of a new type of snacks. The company claims that the mean mass of each packet of snacks is 100 g. After packaging, the mass of 60 packets of the snacks, each of mass x g, is weighed and it is found that $\sum x = 5930$ and $\sum x^2 = 659320$.
- (i) Calculate the unbiased estimates of the population mean and the population variance of the mass of the contents in each packet of the snacks, leaving your answer correct to 3 decimal places. [3]
 - (ii) Find the probability that the mean mass of the 60 packets of snacks is more than 101 g. [3]
 - (iii) Test at the 5% level of significance whether the company's claim, that the mean mass of each packet is 100 g, is overstated. [5]
 - (iv) Using only the information in part (iii) and giving a reason, state whether there is sufficient or insufficient evidence at the 10% level of significance to conclude that the mean mass is less than 100 g. [2]

End of Paper